

TABLE I: (LP bound) Upper bound on the number of classical bits m in any $[[n, k; m, d]]_2$ hybrid stabilizer code with fixed length $n \leq 14$ and dimension k for distance $d = 3, 4, 5$. For $k = 0$, we list the largest dimension of a classical linear binary code. Note that there is, e. g., no stabilizer code $[[13, 5, 4]]_2$, excluding the corresponding entry in the table.

$d = 3$										
$n \backslash k$	0	1	2	3	4	5	6	7	8	
5	2	0	–	–	–	–	–	–	–	
6	3	0	–	–	–	–	–	–	–	
7	4	2	–	–	–	–	–	–	–	
8	4	3	1	0	–	–	–	–	–	
9	5	4	3	1	–	–	–	–	–	
10	6	5	4	2	1	–	–	–	–	
11	7	6	5	4	2	0	–	–	–	
12	8	7	6	5	3	2	0	–	–	
13	9	8	7	5	5	3	1	0	–	
14	10	9	8	7	6	5	3	1	0	

$d = 4$										
$n \backslash k$	0	1	2	3	4	5	6			
5	1	–	–	–	–	–	–			
6	2	–	–	–	–	–	–			
7	3	–	–	–	–	–	–			
8	4	–	–	–	–	–	–			
9	4	–	–	–	–	–	–			
10	5	3	1	–	–	–	–			
11	6	4	2	–	–	–	–			
12	7	5	4	2	0	–	–			
13	8	6	5	4	2	0*	–			
14	9	6	6	5	3	2	0			

$d = 5$										
$n \backslash k$	0	1	2	3						
5	1	–	–	–						
6	1	–	–	–						
7	1	–	–	–						
8	2	–	–	–						
9	2	–	–	–						
10	3	–	–	–						
11	4	0	–	–						
12	4	2	–	–						
13	5	4	–	–						
14	6	5	3	1						

$d = 3$										
$n \backslash k$	0	1	2	3	4	5	6	7	8	
5	2	0	–	–	–	–	–	–	–	
6	3	0	–	–	–	–	–	–	–	
7	4	2	–	–	–	–	–	–	–	
8	4	3	1	0	–	–	–	–	–	
9	5	4	3	1	–	–	–	–	–	
10	6	5	4	2	0	–	–	–	–	
11	7	6	5	4	2	0	–	–	–	
12	8	7	6	5	3	1	0	–	–	
13	9	8	7	6	5	3	1	0	–	
14	10	9	7	7	6	4	2	0	0	

$d = 4$										
$n \backslash k$	0	1	2	3	4	5	6			
5	1	–	–	–	–	–	–			
6	2	–	–	–	–	–	–			
7	3	–	–	–	–	–	–			
8	4	–	–	–	–	–	–			
9	4	–	–	–	–	–	–			
10	5	3	0	–	–	–	–			
11	6	4	2	–	–	–	–			
12	7	5	4	2	0	–	–			
13	8	6	5	3	1	0*	–			
14	9	7	6	5	2	1	0			

$d = 5$										
$n \backslash k$	0	1	2	3						
5	1	–	–	–						
6	1	–	–	–						
7	1	–	–	–						
8	2	–	–	–						
9	2	–	–	–						
10	3	–	–	–						
11	4	0	–	–						
12	4	2	–	–						
13	5	3	–	–						
14	6	4	3	0						

When a hybrid stabilizer code $[[n, k; m, d]]_q$ exists, the linear program for the variables $B_w^\perp$, $A_w^\perp$, A_w , and B_w given by (23)–(29) has an integer solution. For qubit codes, we can strengthen the linear program by additionally considering the shadow enumerator [?]

$$\mathcal{S}_{C_0}(X, Y) = \frac{1}{|C_0|} \mathcal{W}_{C_0}(X + (q^2 - 1)Y, Y - X), \quad (30)$$

which has to have non-negative integer coefficients.

Using CPLEX V12.6.3.0, we checked whether the integer program is feasible. More precisely, we first fix the length n , number of qudits k , and number $M = 2^m$ of classical symbols. Then we look for the largest minimum distance d for which the integer program is found to be feasible. The resulting bounds on the parameters $[[n, k; m, d]]_2$ are listed in Table I, i. e., for fixed parameters n , k , and d , the largest possible value for m is given. For $n > 14$, there seem to be some precision issues, so we list only the bounds for $n \leq 14$.

VI. RESULTS

Based on the construction discussed in Section IV, we perform a search for $\mathcal{C} = [[n, k; m, d]]_2$ codes with distance $d \geq 3$. We start with the self-dual codes from the classification in [?], [?]. In a first step, we construct impure quantum codes $[[n, 1, d]]_2$, and then look for additional vectors for the encoding of classical information, resulting in an $[[n, 1; m', d]]_2$ hybrid code. In some cases it turns out that we can in fact encode more than one qubit, i. e., the code $[[n, 1; m', d]]_2$ is in fact a hybrid code with parameters $[[n, k; m' - k + 1, d]]_2$.

For $d = 4$ and $d = 5$, we have exhaustively searched using all self-dual codes listed in [?], [?] up to $n = 11$. For $d = 3$, we have exhaustively searched all self-dual codes listed in [?], [?] up to $n = 10$. We also have conducted randomized search for $n = 11$. Finally, we appended some qubits in the state $|0\rangle$ to good quantum codes and found new hybrid codes. The results are summarized as follows.

Theorem 6: There exist hybrid codes with the following parameters:

$$\begin{aligned} &[[7, 1; 1, 3]]_2, \quad [[9, 2; 2, 3]]_2, \quad [[10, 3; 2, 3]]_2, \quad [[11, 4; 2, 3]]_2, \\ &[[11, 1; 2, 4]]_2, \quad [[13, 1; 4, 4]]_2, \\ &[[13, 1; 1, 5]]_2, \quad [[14, 1; 2, 5]]_2, \quad [[15, 1; 3, 5]]_2, \\ &[[19, 9; 1, 4]]_2, \quad [[20, 9; 2, 4]]_2, \quad [[21, 9; 3, 4]]_2, \quad [[22, 9; 4, 4]]_2. \end{aligned}$$

All these codes have better parameters than codes obtained from the best quantum codes using Lemma 2.

Below, we provide more details on these codes. In presenting each $[[n, k; m, d]]_2$ code, we first list the generators of the stabilizer of the corresponding $[[n, k, d]]_2$ impure quantum code $\mathcal{C}^{(0)}$, with its $2k$ logical operators between a single and a double horizontal line. The stabilizer of the code $\mathcal{C}^{(0)}$, corresponding to the code C_0 , is generated by the rows above the single horizontal line, while the normalizer of the code $\mathcal{C}^{(0)}$, corresponding to the symplectic dual code C_0^* , is generated by the rows above the double horizontal line. Below the double horizontal line, we list the additional generators that are used to encode m classical bits.